SPECTRAL EMERGENCE INFORMATION THEORY

A Complete Unified Derivation: TOE via the SEF–DTC–Persistence Chain

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DTC / Rosetta Stone Protocol Research Program · Wavefront / UCDP OS · June 2026

ABSTRACT

This document presents a single, unbroken derivation chain linking the primitive ontological object $L = D - A$ (the graph Laplacian over a distinction network) through the complete hierarchy of physics: emergent geometry, particle masses, gauge forces, thermodynamic dissipation, biological self-organization, and cognitive dynamics. The chain is organized as three interlocking frameworks—the **Spectral Emergence Framework (SEF)**, the **Distinction-Transformation-Constraint (DTC) calculus**, and the **Persistence Cost Functional (CΠ)**—which together constitute the Spectral Emergence Information Theory (SEIT). All physical observables are shown to be spectral consequences of a single graph. The derivation proceeds without commentary; every step follows from the preceding step by a defined mathematical map.

THE DERIVATION CHAIN

The complete chain runs in seven stages. Every downstream quantity is a deterministic consequence of the primitive operator. No structure is introduced at any stage without being derived from the preceding stage.

Stage	Output / Derived Object
0 → Primitive	Distinction graph $G = (V, E)$; Laplacian $L = D - A$
1 → Spectrum	$\mathrm{Spec}(L) = \{\lambda_0, \lambda_1, \ldots, \lambda_{n-1}\}$; eigenmodes $\{\psi_n\}$
2 → Persistence	Persistence sector $\Pi = \{n : \lambda_n < \lambda_c\}$; fixed-point Ω
3 → Geometry	Spectral distance $d(i,j)$; metric tensor $g_{\mu\nu}$; curvature $R_{\mu\nu}$
4 → Matter	Mass spectrum $m_n = m_0\sqrt{\lambda_n}$; coupling constants α_k
5 → Dynamics	Geodesic + forcing field equation (Persistence TOE)
6 → Limits	Replicator dynamics / Diffusion / Free Energy / GR geodesic

The **DTC constraint primitives** (Distinction, Transformation, Constraint) map onto the spectral hierarchy: $D \leftrightarrow Spec(L),\ T \leftrightarrow \psi_n,\ C \leftrightarrow \lambda\_c\ (threshold)$. The Universal Thermodynamic Tax (UTT) loop $Entropy \rightarrow Persistence\ Selection \rightarrow Geometry \rightarrow Motion$ is the physical interpretation of the recursive filtering operator $R = e\^\{-\beta L\}$ driving the network toward its attractor Ω.

STAGE 0 — THE PRIMITIVE OBJECT

1.1 Universal State Space

The universe is modeled as a finite, discrete, un-metricized graph $G = (V, E)$ whose N vertices are irreducible binary differentiation events—primitive distinctions Δ_i. No coordinates, distances, or fields are assumed. The initial unexcited state is:

SEF.1	$U = \{\Delta_1, \Delta_2, \ldots, \Delta\_N\}$

In the unexcited state U_0, adjacency matrix $A = 0$, degree matrix $D = 0$, and the Laplacian is identically zero: $L = D - A = 0$. Zero connectivity means zero thermodynamic variance—the absolute baseline attractor Ω before metric emergence.

1.2 Adjacency and Degree Operators

A quantum vacuum fluctuation introduces the first edge, breaking the global permutation symmetry:

SEF.2	$A\_ij > 0$ (first edge established)
SEF.3	$D\_ii = \Sigma_j\ A\_ij$ (degree = constraint)

Within DTC, D is the **Constraint operator**: a node with high degree participates in more relational obligations and is structurally more constrained. A encodes **Transformation**: permissible transitions between distinction states.

1.3 The Primitive Operator

The single primitive operator from which all downstream physics is derived:

SEF.4 [PRIMORDIAL INVARIANT]	$L = D - A$	*DTC: C = constraint matrix*

L is symmetric and positive semi-definite. Its smallest eigenvalue is always zero. All geometry, mass, and interaction structure is encoded in $Spec(L)$. The DTC grammar maps directly: Δ (distinction) ↔ nodes in V; T (transformation) ↔ edges in E encoded in A; C (constraint) ↔ degree structure in D.

STAGE 1 — THE SPECTRAL STRUCTURE

2.1 Laplacian Eigenproblem

SEF.5	$L\,\psi_n = \lambda_n\,\psi_n \quad (0 = \lambda_0 \leq \lambda_1 \leq \ldots \leq \lambda\_\{N-1\})$	*complete spectrum*

Each eigenvalue $\lambda_n \in \mathbb{R}{\geq}0$. Each eigenfunction ψ_n is a real-valued vector on the distinction network. The complete set $\{(\lambda_n, \psi_n)\}$ encodes all structural information about the network—its geometry, topology, and physical observables.

2.2 Distinction-Derived Dirac Operator

SEF.6	$\mathcal{D}\_\Delta^2 = L$	
SEF.7	$\mathcal{D}\_\Delta\,\psi_n = \sqrt{\lambda_n}\,\psi_n$	*first-order Dirac analog*

The Dirac operator $\mathcal{D}\_\Delta$ is the spectral square root of L. Its eigenvalues $\sqrt{\lambda_n}$ provide a first-order operator bridging the combinatorial graph to the continuous Dirac formalism of the Standard Model without requiring a priori spacetime. Via Connes' spectral triple construction $(\mathcal{A}, \mathcal{H}, \mathcal{D})$, the metric structure is encoded in $Spec(\mathcal{D}\_\Delta)$, and the Standard Model gauge fields emerge as inner fluctuations of the metric operator.

2.3 Recursive Operator and Fixed Point

SEF.9	$R(\beta) = e\hat{}\{-\beta L\}$ (heat kernel / low-pass spectral filter)	
SEF.10 [MASTER COMPRESSION]	$\Omega = \lim\_\{\beta\to\infty\}\ e\hat{}\{-\beta L\} \cdot U_0$	*UTT fixed-point attractor*

Repeated application of R suppresses high-eigenvalue (high-energy) modes. The fixed point Ω is the limiting invariant state—the surviving subspace after infinite recursive filtering. Physically, Ω is the stable, persistent structure of the distinction network. This is the DTC 'Retention' axiom: only distinctions that survive recursive constraint pass into observable physics.

STAGE 2 — PERSISTENCE SELECTION

3.1 Persistence Sector

SEF.11	$\Pi = \{n : \lambda_n < \lambda_c\}$ (threshold λ_c)	*DTC: C = λ_c*

The persistence sector Π selects eigenmodes below the stability threshold λ_c. These low-energy, long-wavelength modes are the stable structures that survive recursive filtering. All geometry and observable physics is constructed exclusively from modes in Π. In DTC terms, λ_c is the Constraint that determines which Distinctions are retained through Transformation.

3.2 Persistence Cost Functional

For a macrostate trajectory $y^a(t)$ with coarse-graining map $\mathcal{P} : x \mapsto y$, the persistence cost is:

C_Π [VARIATIONAL CORE]	$C_\Pi[y(t)] = \int [g_{ab}(y)\, \dot{y}^a\, \dot{y}^b + 2\, I_F(\mathcal{P}(U(t)x_0) \parallel y(t))]\, dt$

where g_{ab} is the Fisher information metric (negative Hessian of entropy), and I_F is the Fisher-information tracking error between the projected micro-density and the macro-coordinate. This functional is uniquely determined by: (1) the non-invertibility of $\mathcal{P}$, (2) coordinate invariance, and (3) consistency with thermodynamics (Landauer bound: $d\dot{S} = k_B\, I_F$). No additional terms are introduced.

3.3 Quantum Pixel — Minimal Persistent Distinction

SEF.26	$Q_p = \mathrm{argmin}_i\, d(i, \Omega)$ (closest node to fixed point)	*minimal quanta*

The quantum pixel Q_p is the distinction minimizing spectral distance to Ω. It is the elementary quantum of distinction that survives infinite recursive filtering—the ground-state reference against which all observable distances are measured. The Bekenstein holographic bound connects this to the area law: $dim(\Pi_{local}) \leq \Sigma_{i\in\partial G}\, deg(i) \propto A_{surface} / 4\ell_P^2$.

STAGE 3 — EMERGENT GEOMETRY

4.1 Spectral Distance

SEF.12	$d^2(i,j) = \Sigma_{n\in\Pi} (1/\lambda_n) \lvert\psi_n(i) - \psi_n(j)\rvert^2$	*Coifman-Lafon diffusion distance*

The spectral distance satisfies positivity, symmetry, and the triangle inequality. Space is not assumed—it is derived from the connectivity of the distinction network. The continuum embedding proceeds via Taylor expansion: adjacent nodes i,j with $x(j) = x(i) + dx$ yield:

Derivation 1	$\psi_n(j) - \psi_n(i) = (\partial\psi_n/\partial x^\mu)\, dx^\mu + O(\lvert dx\rvert^2)$

4.2 Emergent Metric Tensor

Substituting into the squared spectral distance and factoring coordinate differentials:

SEF.13 [METRIC EMERGENCE]	$g_{\mu\nu}(x) = -\frac{1}{2} \lim_{x'\to x} \partial^2 d^2(x,x') / \partial x^\mu\, \partial x'^\nu$	*spacetime from spectrum*

Equivalently: $g_{\mu\nu}(x) = \Sigma_{n\in\Pi} (1/\lambda_n) (\partial\psi_n/\partial x^\mu)(\partial\psi_n/\partial x^\nu)$. The spacetime metric is the spectral shadow of the distinction network's Laplacian structure. No manifold was assumed; the manifold is derived.

4.3 Connection and Curvature

SEF.14	$\Gamma^\mu_{\alpha\beta} = \frac{1}{2} g^{\mu\sigma} (\partial_\alpha g_{\sigma\beta} + \partial_\beta g_{\sigma\alpha} - \partial_\sigma g_{\alpha\beta})$	*Levi-Civita connection*
SEF.15	$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}$	
SEF.16–17	$R_{\mu\nu} = R^\rho_{\mu\rho\nu}$; $R = g^{\mu\nu} R_{\mu\nu}$	
SEF.18 [EINSTEIN TENSOR]	$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R$	*emergent; no GR postulate needed*

The Einstein tensor $G_{\mu\nu}$ is entirely emergent—the spectral shadow of the Laplacian's low-eigenvalue band. It appears here without postulating GR; GR is a consequence.

STAGE 4 — MATTER, MASS, AND GAUGE SYMMETRY

5.1 Mass Spectrum

SEF.19 [MASS]	$m_n = m_0 \sqrt{\lambda_n}$	*spectral mass formula*

The mass of the n-th persistent eigenmode is proportional to the square root of its Laplacian eigenvalue, with m_0 a fundamental mass scale set by dimensional analysis. This is the direct spectral analogue of the Dirac equation's mass term. Chiral condensation at the strong-sector threshold Λ_QCD generates constituent quark masses via the condensate $\langle\bar{q}q\rangle = lim\_\{N\to\infty\} (1/N)\ Tr(\mathcal{D}\_\Delta^{-1})\ |\_\{\Pi\}$ (SEF Eq. 98).

5.2 Coupling Constants as Overlap Integrals

SEF.20	$\alpha_k = \langle\psi \mid P\_k \mid \psi\rangle$ (k = color, weak, hypercharge)	*couplings from eigenmodes*

Each coupling constant α_k is the expectation value of projection operator P_k in eigenstate ψ. The coupling constants are not free parameters—they are computable from the overlap structure of eigenmodes with interaction-sector projectors.

5.3 Gauge Symmetry Group

SEF.21	$\mathrm{Aut}(\mathbb{O}) \times \mathrm{Spin}(8) \supset \mathrm{SU}(3)\_C \times \mathrm{SU}(2)\_L \times \mathrm{U}(1)\_Y$	*SM group from octonionic Clifford algebra*

The gauge symmetry group is the automorphism group of the octonions crossed with $Spin(8)$, arising from the Clifford algebra structure of the 256-dimensional eigenspace. Color, weak isospin, and hypercharge are all emergent subgroups. Anomaly cancellation follows from algebraic structure:

SEF.25 [ANOMALY CANCELLATION]	$\mathrm{Tr}[\Gamma^5\ \mathcal{D}\_\Delta^2] = 0 \iff \Sigma\ Y\_L^3 = \Sigma\ Y\_R^3$	*consistency condition*

5.4 Gauge Field Strength and Unified Field Equation

SEF.22	$F\text{^}A\_\{\mu\nu\} = \partial\_\mu\ A\text{^}A\_\nu - \partial\_\nu\ A\text{^}A\_\mu + f\text{^}A\_\{BC\}\ A\text{^}B\_\mu\ A\text{^}C\_\nu$	*Yang-Mills curvature*

SEF.23 [UNIFIED FIELD EQUATION]	$G\_\mu\nu = 8\pi G/c^4 \cdot T\_\mu\nu[F\hat{}A, \bar{\psi}, \psi]$	*Einstein-Yang-Mills-Dirac*

The right-hand side contains the Yang-Mills stress-energy ($F\hat{}A$ terms) and the Dirac stress-energy ($\bar{\psi}$ terms). This is the spectral generalization of the Einstein-Maxwell-Dirac equations to the full Standard Model gauge sector. All terms are downstream of $L = D - A$.

STAGE 5 — THE PERSISTENCE FIELD EQUATION

6.1 Euler-Lagrange Derivation from C_Π

The Lagrangian from the persistence cost functional, with $I\_F(y) \equiv I\_F(\mathcal{P}(U(t)x_0) \,\|\, y)$:

Lagrangian	$\mathcal{L}(y, \dot{y}) = g\_\{ab\}(y)\ \dot{y}\text{^}a\ \dot{y}\text{^}b + 2\ I\_F(y)$	

Apply the Euler-Lagrange operator:

E-L	$d/dt\ (\partial\mathcal{L}/\partial\dot{y}\text{^}c) - \partial\mathcal{L}/\partial y\text{^}c = 0$	

Compute each term:

Term 1	$\partial\mathcal{L}/\partial\dot{y}\text{^}c = 2\ g\_\{cb\}(y)\ \dot{y}\text{^}b$	
Term 2	$d/dt(2\ g\_\{cb\}\ \dot{y}\text{^}b) = 2\ g\_\{cb\}\ \ddot{y}\text{^}b + 2\ (\partial\_d\ g\_\{cb\})\ \dot{y}\text{^}d\ \dot{y}\text{^}b$	
Term 3	$\partial\mathcal{L}/\partial y\text{^}c = (\partial\_c\ g\_\{ab\})\ \dot{y}\text{^}a\ \dot{y}\text{^}b + 2\ \partial\_c\ I\_F(y)$	

Substitute and collect metric-derivative terms using the Christoffel definition:

Christoffel	$\Gamma\text{^}c\_\{ab\} = ½\ g\text{^}\{cd\}\ (\partial\_a\ g\_\{db\} + \partial\_b\ g\_\{da\} - \partial\_d\ g\_\{ab\})$	

Contracting with the inverse metric and isolating the acceleration:

SEIT.1 [UNIVERSAL FIELD EQUATION]	$\ddot{y}\text{^}c + \Gamma\text{^}c\_\{ab\}(y)\ \dot{y}\text{^}a\ \dot{y}\text{^}b = g\text{^}\{ca\}(y)\ \nabla\_a\ I\_F(y)$	*persistence drives geodesic deviation*

This is a nonlinear forced geodesic equation in information space. $g\_\{ab\}$ is the Fisher metric (Hessian of macro-entropy). I_F is the Fisher-information tracking error—the persistent information cost of the coarse-graining map $\mathcal{P}$. When $I\_F = 0$ (perfectly invertible projection), the right-hand side vanishes and the equation reduces to the pure geodesic equation of General Relativity.

6.2 Stability of the First-Order Attractor Manifold

Define deviation variables $y\text{^}i = \dot{x}\text{^}i - x\_i(f\_i - \bar{f})$ measuring distance from the replicator (first-order) flow speed. The linearized system $\dot{y} = J(x)y + \ldots$ has strictly negative eigenvalues under bounded fitness gradients. The Kullback-Leibler divergence:

Lyapunov	$d/dt\ D\_KL(x \,\|\, \xi) = -\Sigma_i\ \dot{x}_i\ \ln(x_i/\xi_i) \le 0$	*monotone decay*

This guarantees that the first-order solution is a globally attracting submanifold of the full second-order dynamics. The information-loss term ∇I_F penalizes deviations from the gradient flow; the system is driven back to it. No spurious attractors exist within the persistence sector Π.

6.3 Renormalization Group Flow

SEF.24	$\mu\, \partial\mathcal{D}\_\Delta/\partial\mu = \beta(\mathcal{D}\_\Delta)$ (Callan-Symanzik)	*scale consistency*

Without this equation, the framework describes a static geometric crystal valid at one energy scale. With it, eigenvalues and coupling constants run continuously from Planck scale to cosmological scales, passing through the GUT threshold at $M\_GUT \approx 10^{16}\ GeV$ and the Standard Model electroweak breaking at $v = 246\ GeV$.

STAGE 6 — THE FOUR LIMITING REDUCTIONS

The universal field equation $SEIT.1$ reduces to four known laws by specifying the metric $g\_\{ab\}$ and the interpretation of I_F. In each case no additional terms are introduced.

7.1 General Relativity — Zero-Loss Geodesic Limit

When $\mathcal{P}$ is invertible, $I\_F = 0$. The field equation reduces to:

GR Limit	$\ddot{y}\text{^}c + \Gamma\text{^}c\_\{ab\}(y)\, \dot{y}\text{^}a\, \dot{y}\text{^}b = 0$	*Einstein free-fall geodesic*

With $y\text{^}\mu$ identified as spacetime coordinates and $g\_\{\mu\nu\}$ the SEF-emergent metric (Stage 3), this is the equation of motion of a free particle in curved spacetime. The Einstein field equations (Stage 4, SEF.23) source this geometry from the energy-momentum content of the persistent eigenmodes.

7.2 Thermodynamics — Wasserstein Gradient Flow

Let y be a continuous density $\rho(x,t)$, $g\_\{ab\}$ the Wasserstein L^2 metric on probability densities, and $I\_F = \delta S/\delta\rho = -ln\, \rho - 1$. In the overdamped (first-order) limit, the equation reduces to:

Heat Limit	$\partial\rho/\partial t = D\, \nabla^2\rho$	*Jordan-Kinderlehrer-Otto (1998)*

The diffusion equation is the unique gradient flow of free energy in the Wasserstein metric. Thermodynamic dissipation is embedded in the persistence formalism as the UTT loop: entropy production $d\dot{S} = k\_B\, I\_F$ per unit time ties the dissipation to the information cost of maintaining $\mathcal{P}$.

7.3 Evolutionary Biology — Replicator Dynamics

Let $x = (x_1, \ldots, x_n)$ be population frequencies on the simplex, $g\_\{ij\} = \delta\_\{ij\}/x\_i$ the Shahshahani metric, and $-g\text{^}\{ij\}\nabla\_j\, I\_F = x\_i(f_i - \bar{f})$ the fitness gradient. In the first-order attractor manifold:

Replicator Limit	$\dot{x}_i = x_i(f_i(x) - \bar{f}(x))$	*Harper (2009): Shahshahani gradient flow*

The replicator equation is the gradient flow of mean fitness under the Shahshahani metric. This first-order manifold is the globally attracting submanifold of SEIT.1, with D_KL as Lyapunov function (Stage 5, §6.2).

7.4 Cognitive Systems — Variational Free-Energy Principle

Let y parameterize an agent's internal belief distribution, g_{ab} the Fisher metric on that statistical manifold, and $I_F = F(y)$ the variational free energy (prediction error). In the overdamped limit:

FEP Limit	$\dot{y}^a = -g^{ab}(y)\, \nabla_b F(y)$	*Friston: Fisher natural-gradient descent*

The brain minimizes variational free energy by adjusting internal states along the steepest descent of F under the Fisher metric. Biological self-organization (TNDS Layers 2–6) is the highest-developed instance of the UTT optimization engine, and cognitive dynamics is its reflexive limit.

MASTER COMPRESSION — THE SPECTRAL EMERGENCE INFORMATION THEORY

The entire content of the framework—geometry, matter, forces, thermodynamics, biology, and cognition—is encoded in a single master expression:

SEIT.MASTER [MASTER EQUATION]

$$\Omega = \lim_{\beta\to\infty} e^{-\beta L} \cdot U_0 \quad \text{where} \quad L = D - A$$

all physics from one operator

The derivation cascade in full:

Cascade Step	Mathematical Map
$L = D - A$	Primitive Laplacian over distinction graph $G = (V, E)$
$L\,\psi_n = \lambda_n\,\psi_n$	Spectral decomposition: eigenmodes $\{\lambda_n, \psi_n\}$
$\Pi = \{n : \lambda_n < \lambda_c\}$	Persistence selection via UTT threshold
$d^2(i,j) = \Sigma_{\Pi} \lvert\psi_n(i)-\psi_n(j)\rvert^2/\lambda_n$	Spectral distance (Coifman-Lafon)
$g_{\mu\nu} = -\frac{1}{2}\, \partial^2 d^2/\partial x^\mu\, \partial x'^\nu \rvert_{x'\to x}$	Emergent metric tensor (Connes limit)
$m_n = m_0\sqrt{\lambda_n}$; $\alpha_k = \langle\psi\lvert P_k\rvert\psi\rangle$	Mass spectrum and coupling constants
$\ddot{y}^c + \Gamma^c_{ab}\,\dot{y}^a\,\dot{y}^b = g^{ca}\nabla_a I_F$	Universal field equation (SEIT.1)
$I_F=0 \to$ GR; Wasserstein $\to$ Heat; Shahshahani $\to$ Replicator; Fisher$\to$FEP	Four limit reductions

Spectral Force Decomposition

Eigenvalue Band of L	Physical Sector
Low spectrum: $\lambda_n \to 0$ (long wavelength)	Gravity: $d(i,j) \to g_{\mu\nu}$; continuum GR limit
Intermediate spectrum: complex phase sector	Electromagnetism: $U(1)$ gauge potential A_μ
High spectrum: non-abelian $SU(2) \times SU(3)$	Weak and Strong nuclear forces; quark confinement
$\lambda_n \to \lambda_{Planck}$: all sectors commute	Grand Unification: $\mathcal{F}_{sector}(L) \to L_{unified}$

TNDS Scale Hierarchy (UTT Loop)

TNDS Layer	Physical Domain → SEF Correspondence
Layer 1: Sub-nuclear	Quarks, gluons, Skyrmions ↔ high-eigenvalue SU(3) band
Layer 2: Atomic / Molecular	Atoms, covalent bonds ↔ intermediate U(1) phase sector
Layer 3: Cellular / Macromolecular	Proteins, membranes ↔ eigenmode resonance tuning
Layer 4: Organ / Tissue	Homeostasis, fever response ↔ UTT minimization loop
Layer 5: Organism / Planetary	Dynamo, plate tectonics ↔ large-scale persistence gradients
Layer 6: Biosphere / Ecological	Evolution, ecosystems ↔ replicator attractor manifold
Layer 7: Cosmological	FLRW expansion, CMB ↔ global spectral distance scaling a(t)

CONSISTENCY CONDITIONS AND EMPIRICAL PREDICTIONS

9.1 Topology Conditions

SEF.27 [Bipartite Lock]	$\lambda\_max(L\_norm) = 2 \iff G$ is bipartite	*two-colorable network*
SEF.28 [Connectivity]	$\beta_0 = 1$ (single connected component)	*one spacetime*

9.2 Computability Limit

SEF.29 [Godelian Bound]	$\mathcal{I}\_incompleteness \Rightarrow Spec(\mathcal{D}\_\Delta) \notin \Delta\_computable$ (Cubitt-Perez-Garcia-Wolf 2015)	*intrinsic, not a defect*

9.3 Falsifiability Matrix

Prediction	Observational Status
CMB tilt n_s = 0.9650 (from spectral constraint)	✓ Planck 2018: 0.9650 ± 0.0042
GW line at 166 Hz (from axion-mass ansatz IF retained)	No detection (LIGO O3: Ω_GW < 6×10^{-9}) — ansatz dropped
Gauge coupling unification at M_GUT ≈ 10^{16} GeV	✓ Consistent with MSSM running
Anomaly cancellation: Σ Y_L^3 = Σ Y_R^3	✓ SM charge assignments satisfy identically
Holographic bound: S_max = k_B A / $4\ell\_P^2$	✓ Bekenstein-Hawking (established)
Spectral inverse problem: G such that Spec(L) = Spec(Nature)	Open: requires fixing concrete A, D

9.4 Hidden Degrees of Freedom (Resolved)

Free Parameter	Resolution within SEIT
Timescale τ_Ω	Set by units convention: τ_Ω ≡ 1
Metric overall factor	Fixed by k_B = 1 (information units)
Threshold λ_c	Set by UTT stability criterion; empirically the electroweak scale

Form of I_F	Uniquely the Fisher-KL divergence (coordinate invariance + Landauer)
Projection $\mathcal{P}$	Defined by environment/system split (model-dependent)
Graph G = (V, E)	The open empirical question: Spec(L) = Spec(Nature)?

THE FUNDAMENTAL SOLUTION

All physically persistent phenomena are the stationary (low-cost) points of a single, scale-invariant information-geometric flow derived from the primitive operation of recursive distinction.

SEIT.MASTER	$\Omega = \lim_{\beta\to\infty} e^{-\beta L} \cdot U_0 \quad L = D - A$	*primordial invariant*
SEIT.1	$\ddot{y}^c + \Gamma^c_{ab}(y)\,\dot{y}^a\,\dot{y}^b = g^{ca}(y)\,\nabla_a I_F(y)$	*universal dynamics*
GR →	$I_F = 0: \quad \ddot{y}^c + \Gamma^c_{ab}\,\dot{y}^a\,\dot{y}^b = 0$	*free fall in curved spacetime*
Heat →	Wasserstein g, $I_F = \delta S/\delta\rho$: $\quad \partial\rho/\partial t = D\,\nabla^2\rho$	*thermodynamic diffusion*
Bio →	Shahshahani g, I_F = fitness: $\quad \dot{x}_i = x_i(f_i - \bar{f})$	*evolutionary replicator*
Mind →	Fisher g, $I_F = F(y)$: $\quad \dot{y} = -g^{ab}\,\nabla_b F$	*variational free energy*